

RAJ KUMAR

Lead Backend Engineer & Systems Architect

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PROFESSIONAL SUMMARY

Accomplished **Lead Backend Engineer & Systems Architect** with **8+ years of experience** designing, scaling, and maintaining resilient distributed architectures, asynchronous message pipelines, and high-concurrency microservices. Core expertise in **Python 3.12 (FastAPI, AsyncIO)**, **TypeScript**, **Node.js**, **NestJS**, and **Go/PHP**, with deep specialization in enterprise middleware: multi-tenant **X12 EDI & AS2 cryptographic transports**, **Xero accounting/payroll sync**, real-time **Socket.IO & WebSockets**, and provider-agnostic **Generative AI model orchestration** (Gemini, ComfyUI, OpenAI). Expert in transactional integrity (double-entry ledgers, idempotent deduplication), distributed caching (Redis 7), worker queues (BullMQ, ARQ), relational/NoSQL optimization (PostgreSQL asyncpg, MongoDB, MySQL), and Dockerized cloud deployments.

TECHNICAL COMPETENCIES

Backend & Architecture:	NestJS 11, FastAPI (AsyncIO), Node.js, AdonisJS v6, Express.js, Microservices, Event-Driven Systems, REST, WebSockets, Socket.IO
Languages & Runtimes:	Python 3.12, TypeScript, JavaScript (ES6+), PHP, Java, SQL, C++, Bash/Shell scripting
Protocols & Middleware:	X12 EDI (850, 945, 944, 810, 997, 855), AS2 Protocol (OpenSSL CMS Signing & AES-256 Encryption, sync/async MDN), SFTP, S3/SQS
AI & Computer Vision:	Generative AI Orchestration (Gemini Vertex AI, ComfyUI, Decart/FASHN), OpenAI SDK, Computer Vision (OpenCV, K-Means Clustering, Vector SVG)
Databases & Caching:	PostgreSQL 16 (asyncpg/Neon), MongoDB/Mongoose 8, MySQL, Redis 7 (Pub/Sub, Caching, Streams), TypeORM, SQLAlchemy 2.0, Drizzle
Job Queues & Async:	BullMQ, ARQ Workers, Redis Queues, Background Schedulers, Idempotent Webhook Event Store, Asynchronous File Streams
Integrations & Billing:	Xero Accounting/Payroll API, Stripe (Connect Express, BECS Direct Debit, Webhooks), PRODA/CCS Gov Tokens, Mapbox GL
DevOps, Cloud & Security:	Docker Compose, AWS (S3, SQS, Secrets Manager), Azure Blob, JWT + OAuth2 rotation, TOTP 2FA, OpenSSL CMS, Git, Linux

PROFESSIONAL EXPERIENCE

Lead Backend Engineer & Systems Architect — Luminoguru Pvt Ltd 2020 – Present | Mohali, India

- Direct the architectural strategy and backend implementation of enterprise systems, microservices, and asynchronous pipelines using Python (FastAPI), TypeScript (NestJS), and Node.js.
- Engineered production X12 EDI middleware in Python 3.12 with AS2 encryption/signing, custom parser/validator, and ERP integration.
- Built workforce management platform (TGA-Mesh) with Socket.IO messaging, background timesheet workers, and automated Xero payroll sync.
- Architected multi-model Generative AI fitting room orchestration layer routing jobs dynamically across Gemini, ComfyUI, and Decart.
- Designed computer vision image-to-stencil vector engine (StenGen) combining NestJS microservices and Python OpenCV image quantization.
- Constructed high-throughput social travel infrastructure (Globetrotrr) with distributed Redis caching, BullMQ queues, and Mapbox geospatial querying.

Backend & Web Developer — Quadrish Innovations 2019 – 2020 | Chandigarh, India

- Designed and maintained scalable relational database models and RESTful APIs with PHP (CodeIgniter, Laravel) and MySQL.
- Built asynchronous data ingestion services and integrated Node.js and MongoDB micro-modules to optimize real-time throughput.

Software & Mobile Engineer (Trainee to Developer) — Tuespot Pvt Ltd 2017 – 2019 | Mohali, India

- Engineered native Android mobile applications with Java/XML, building local SQLite storage layers, background sync, and REST consumers.
- Contributed to dynamic web portal architectures, backend schema design, and server-side script optimizations with PHP and SQL.

FEATURED BACKEND & SYSTEM ARCHITECTURE PROJECTS

1. TGA-Mesh — Workforce Operations Backend & Mobile App

[Google Play Store Link](#)

Stack: NestJS 11, TypeScript, MongoDB, Mongoose 8, Redis, BullMQ, Socket.IO, Xero API, React Native, Firebase, AWS S3

- Architected multi-centre workforce backend (~371 TS files, 30 controllers, 29 schemas) handling shifts, leave requests, and timesheet worker aggregation.
- Built automated Xero Accounting & Payroll sync service with encrypted bank credentials, AWS Secrets Manager storage, and queued timesheet reconciliation.

- Implemented real-time Socket.IO communication cluster with typing indicators, presence, and GPS-fenced mobile attendance (published on Google Play).

2. EDI Middleware — Standalone Multi-Tenant X12 & AS2 Platform

Enterprise Production Middleware

Stack: Python 3.12, FastAPI (AsyncIO), PostgreSQL 16, SQLAlchemy 2.0, Redis 7, ARQ Workers, AS2 (OpenSSL CMS), pyx12, AWS SQS/S3, Docker

- Built async multi-tenant X12 pipeline: transport → parse → validate → map → ERP → ack/audit, handling 850/945/944/810 docs with 997/855/940 generation.
- Implemented production AS2 transport: OpenSSL CMS signing/encryption (AES-256), partner certificate store, sync/async MDN, and integrated live partner (Solvira).
- Engineered resilient ingestion engine with Redis + ARQ workers, idempotent deduplication on control numbers (409 handling), and swappable REST ERP adapter (Icicle Foods).

3. FashionTryOn — AI Virtual Fitting Room Engine & Shopify Plugin

AI Model Orchestration

Stack: Python, FastAPI, Node.js, Gemini API (Vertex AI), ComfyUI Cloud, Decart/FASHN, Shopify App Bridge, Redis, BullMQ, AWS S3

- Architected provider-independent generative AI fitting room orchestration layer decoupling e-commerce logic from underlying ML diffusion models.
- Built dynamic AI routing adapter across Google Vertex AI / Gemini, ComfyUI workflows, and Decart/FASHN with automated model failover and cost optimization.
- Designed async worker queue with Redis and webhook callbacks for GPU-heavy image inference without blocking merchant storefront execution.

4. StenGen (TattooStencils) — AI Image-to-Stencil Vector Engine

stengen.com

Stack: Python 3.11, OpenCV, K-Means Clustering, NumPy, NestJS, TypeScript, Docker, AWS S3, Vector SVG Engine

- Engineered multi-stage computer vision pipeline: luminance thresholding, K-Means segmentation, bilateral filtering, and automated vector path tracing.
- Built asynchronous micro-service bridge connecting NestJS API gateway with Python CV workers for low-latency image processing and SVG delivery.
- Implemented algorithmic line density and shading controls, supporting instant vector SVG rendering and high-DPI thermal stencil print exports.

5. Globetrottr — Travel Social Network & Real-Time Messaging Platform

globetrottrapi.lusites.xyz/swagger

Stack: TypeScript, NestJS 11, MongoDB, Mongoose 8, Redis, BullMQ, Socket.IO, Mapbox GL, Elasticsearch, Typesense, AWS S3, Stripe

- Architected high-throughput social backend (~338 TS files, 29 controllers, 60 services) powering social travel feeds, stories, and Mapbox clustering.
- Built high-concurrency real-time Socket.IO messaging gateway with typing indicators, presence, AWS S3 media pipeline, and Google Cloud translation.
- Integrated Tiqets Distributor API for travel event discovery, business accounts with QR redemption, and Stripe subscriptions.

ADDITIONAL PRODUCTION SYSTEMS & ENTERPRISE PLATFORMS

Comprehensive documentation, architecture breakdowns, and demos for 15+ production systems viewable at: <https://rajansandha.github.io/dev-portfolio>

- **TGA Grove Academy:** Enterprise childcare management & compliance platform (500+ files, 56 entities) with Australian PRODA/CCS tokens & Stripe BECS. (NestJS, PostgreSQL, PRODA)
- **Talent Bridge:** Full-stack AI recruitment platform with automated resume parsing (PDF/DOCX), OpenAI candidate scoring, and BullMQ queues. (NestJS, OpenAI SDK, Redis)
- **Givicle:** Full-stack CSR employee-giving platform with double-entry contribution ledger, payroll deductions, and statement exports. (AdonisJS v6, MySQL, Redis, BullMQ, Azure Blob)
- **BarChase API:** Nightlife discovery platform with Mapbox geo-filtering, crowd status tracking, Stripe sponsorships, and push notifications. (NestJS 11, MySQL, TypeORM, Stripe, Firebase)
- **Village Vest:** Community group-funding escrow platform with Stripe Connect Express onboarding, OFAC compliance screening, and SMS OTP. (NestJS 11, Stripe Connect, Twilio)
- **Boldify:** High-volume Shopify brand storefront engineered with custom modular Liquid snippet architecture. (Shopify Liquid, Custom Apps)
- **Mobile & Open Source:** ChessRoyal Game (Google Play), ContentCreationAI (Voice cloning), CompileCraftCode, NanoNomad. (Open Source Repositories)

EDUCATION

Master of Computer Applications (MCA)

2016 – 2018 | Grade A

Maharaja Ranjeet Singh Punjab Technical University (MRSPTU), Punjab, India

Bachelor of Computer Applications (BCA)

2012 – 2015 | Grade A (65%)

Panjab University, Chandigarh, India